TIAGO FERRÃO-LINDQVIST

Data Engineering | Test Data Processing & Reporting

PROFESSIONAL SUMMARY

Test and Validation Engineer with experience building Python-based tools to process rig data, check test results against specifications and generate reports. Developed a nightly processing workflow handling approximately 2 GB of TDMS and CSV data per week from six rigs, with results written to a staging table in SQL Server. Also built a Power BI dashboard used by validation and design engineers. Transitioning into data engineering through hands-on Python data processing, scheduled workflows, SQL Server reporting and ongoing data engineering study.

TECHNICAL SKILLS

Programming and data processing: Python scripting, pandas, NumPy, SQL, MATLAB, Simulink

Data and reporting: SQL Server, SQLite, Power BI, Matplotlib, npTDMS

Other tools: Flask, Git (personal projects), Docker Compose (tutorial-based personal project), Windows Task Scheduler

PROFESSIONAL EXPERIENCE

Test & Validation Engineer | Corvane Automotive Components *2019–Present*

- Conduct durability and vibration testing, test-data tooling and root-cause analysis.
- Built Python scripts using pandas, NumPy and npTDMS to parse TDMS and CSV rig output, process approximately 2 GB of data per week across six rigs, validate channels against test specifications, and generate report figures.
- Scheduled a nightly workflow on a lab PC using Windows Task Scheduler; wrote summary results to an IT-created staging table in SQL Server.
- Packaged the tool for internal use; four other test engineers use it for their own tests.
- Created a Power BI dashboard using the staging table to show pass/fail trends and cycles to failure by product generation; used by approximately 15 engineers across validation and design.
- Developed a MATLAB/Simulink model of a seat-adjuster drive, including motor, gearbox and rail friction, to select rig load profiles matching customer duty cycles.
- Trained three apprentices in rig safety and test setup (2022–2024).

Mechanical Design Engineer | Halvik Pumps *2014–2019*

- Designed pump casings and impellers and performed structural finite element analysis (FEA).
- Supported performance test-bench activities.

PROJECTS

Bike-sharing trips API | Personal learning project *2025*

- Built a small Flask API over a public bike-sharing dataset loaded into SQLite, with three read-only endpoints.
- Runs locally on a laptop; followed a tutorial to run it with Docker Compose.
- Technologies: Python, Flask, SQLite, Docker Compose, Git.

PROFESSIONAL DEVELOPMENT

Online Data Engineering Course *In progress*

- Completed 6 of 9 modules; no certificate awarded to date.

ADDITIONAL INFORMATION

Portuguese citizen (EU). Happy to commute to Lisbon for hybrid work.